

Rashed Ahmed Alneyadi

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PROFILE

UAE National mathematics & computer science graduate (NYU Abu Dhabi, Dec 2024) with research experience in robotics path optimisation, quantum simulation, and Python/MATLAB modelling. Seeking AI research engineering, data science, and applied ML roles at UAE's leading technology and space organisations.

EXPERIENCE

Research Intern — MIT Zero Robotics (via MBRSC) — Mohammed Bin Rashid Space Centre Jun – Aug 2022
| Dubai, UAE / MIT, USA

- Competitively selected from UAE applicants for MBRSC-sponsored research placement at MIT; developed graph-theory path optimisation in Python and MATLAB for NASA's Astrobee free-flying robot aboard the ISS.
- Implemented Dijkstra's algorithm with dynamic obstacle avoidance; reduced average traversal time ~30% vs. baseline A* approach in simulation.
- Algorithm competed in the international ISS Zero Robotics Final Tournament — final stage involved live execution aboard the International Space Station.
- Presented technical findings to MBRSC scientists; documented algorithm design and test cases for reproducibility.

Quantum Computation Research Assistant — NYU Abu Dhabi Sep – Dec 2022 | Abu Dhabi, UAE

- Collaborated with faculty to study entanglement dynamics in 3-qubit XXZ Heisenberg spin-chain model using density-matrix renormalisation group methods.
- Automated 50–100 MATLAB parameter sweeps across coupling constants and temperature ranges; reduced compute time ~40% through vectorised batch execution.
- Analysed phase-transition signatures in von Neumann entropy curves; findings contributed to lab's ongoing quantum information research programme.

Active Investments & Equity Intern — Abu Dhabi Investment Council (ADIC) Jun – Aug 2023 | Abu Dhabi, UAE

- Built Python multi-factor attribution model explaining ~70% of return variance across 20+ asset managers — applied PCA and OLS regression on rolling return windows.
- Developed predictive analytics pipeline (pandas, NumPy, scikit-learn) to forecast forward manager performance based on factor exposures and market-regime classification.
- Produced automated weekly reporting dashboards connecting live data feeds to visualisation layer.

Private Equity Intern — Abu Dhabi Investment Authority (ADIA) Jan – May 2023 | Abu Dhabi, UAE

- Applied quantitative screening frameworks to evaluate 40+ APAC deal opportunities; built 12+ financial models in Python and Excel.
- Synthesised data-driven insights into IC memos — bridged quantitative analysis and investment narrative.

CEO & Co-Founder — Polygon Technical Infrastructures 2021 – Present | Abu Dhabi, UAE

- Founded full-stack software and cybersecurity firm; architected and shipped client-facing products using React/TypeScript, Python, and MongoDB with CI/CD pipelines.
- Led penetration testing and incident response engagements; developed bespoke automation scripts for vulnerability scanning.

Co-Founder — RECTify Brokers 2023 – Present | Abu Dhabi, UAE

- Designed data pipeline for emissions baselining and REC/carbon-credit procurement platform; integrated external APIs for real-time certificate pricing.

EDUCATION

BA Mathematics — Minor Computer Science — NYU Abu Dhabi Aug 2021 – Dec 2024 | Abu Dhabi, UAE

- GPA: 3.4 / 4.0
- Relevant coursework: Probability & Statistics, Real Analysis, Algorithms, Machine Learning, Econometrics, Linear Algebra, Discrete Math
- Awards: Inside ADIA Competition — 2nd Place (2023) | Wharton Innovation Competition — 3rd Place (2022)

SKILLS & TOOLS

Machine Learning: scikit-learn, regression, classification, PCA, time-series analysis, parameter optimisation

Scientific Computing: Python (NumPy, pandas, matplotlib, SciPy), MATLAB (vectorised simulation, parameter sweeps)

Algorithms: Graph theory, Dijkstra, dynamic programming, optimisation, Monte Carlo simulation

Engineering: React / TypeScript, MongoDB, Node.js, DevOps, CI/CD, Git, Docker basics

Languages: Python, MATLAB, Java, C, JavaScript / TypeScript

Human Languages: Arabic (native), English (fluent)

INTERESTS

AI for space & defence applications | Autonomous systems & robotics | Quantum machine learning | Geospatial data science | Research engineering